

Cloud Lab: Cross-Region PAM Deployment with VNet Peering

Objective

To deploy a Privileged Access Management (PAM) lab using senhasegura in Microsoft Azure, and securely connect it to both Linux and Windows target systems across regions using VNet Peering, while optimizing for minimal cost.

Lab Architecture

◆ PAM Appliance VM

- **Region:** UAE North
 - **VNet:** vnet-uaenorth
 - **Address Space:** 172.16.0.0/16
-

◆ Linux Target VM

- **Region:** Central India
 - **VNet:** linux-vm-vnet
 - **Address Space:** 10.0.0.0/16
 - **Access:** SSH (Port 22)
-

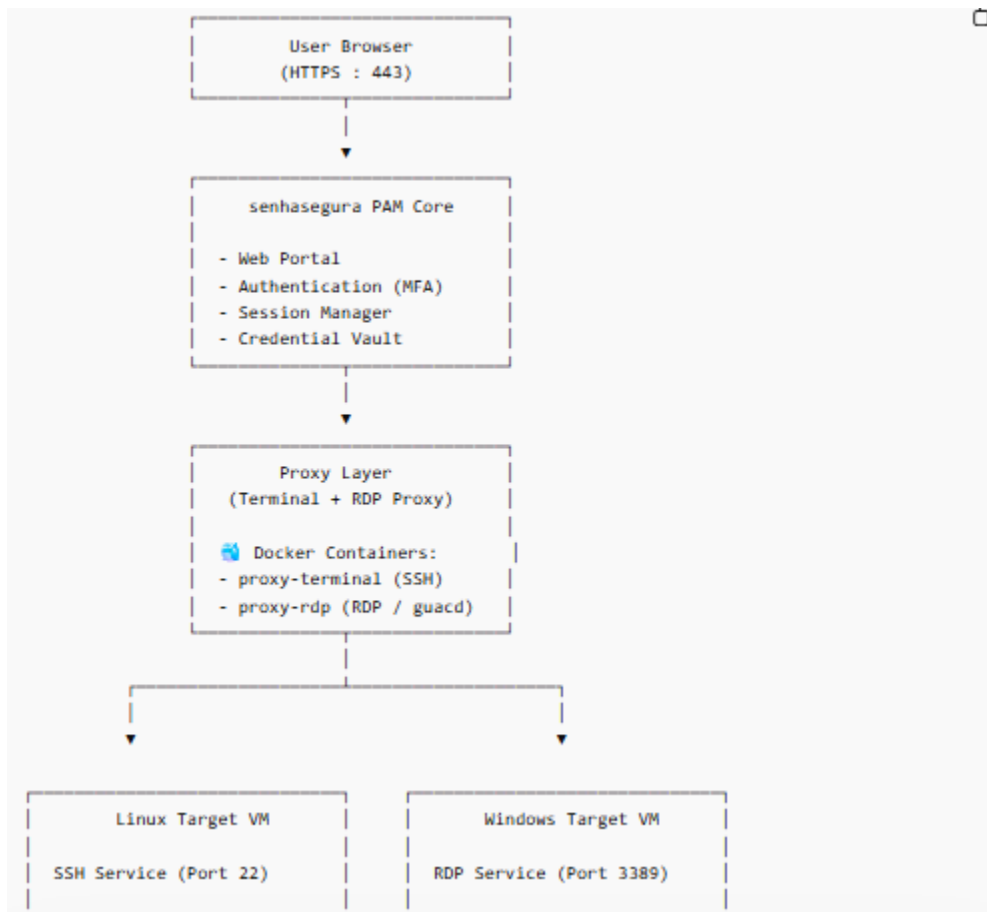
◆ Windows Target VM

- **Region:** Central India
 - **VNet:** linux-vm-vnet (same as Linux VM)
 - **Access:** RDP (Port 3389)
-

◆ Connectivity

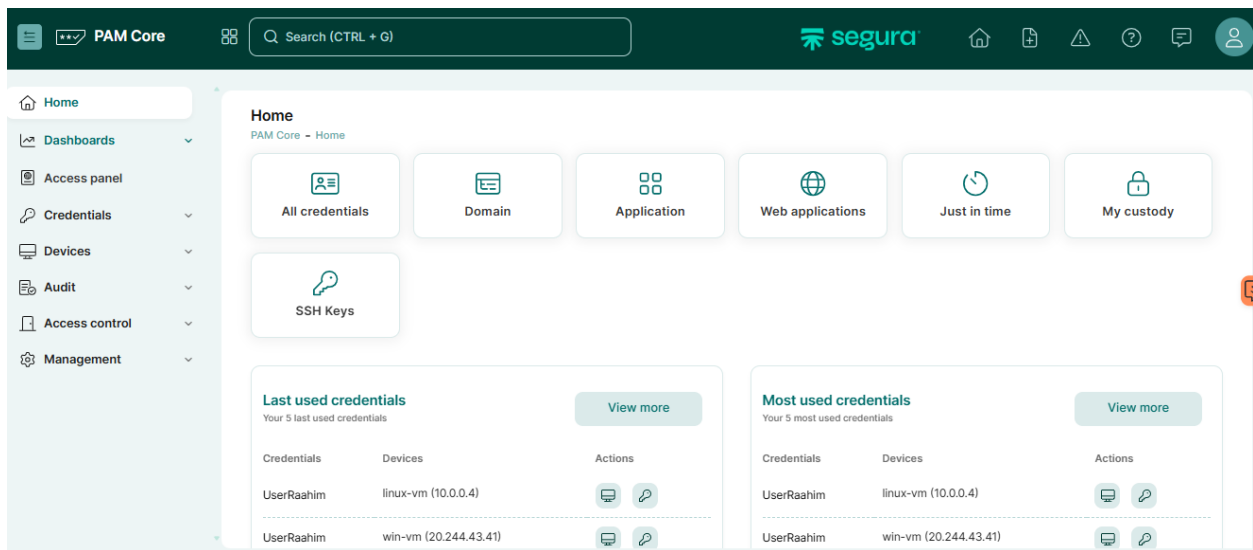
- **Method:** VNet Peering (Azure Private Backbone)

- **Type:** Bidirectional



Step 1: Deploy PAM Appliance

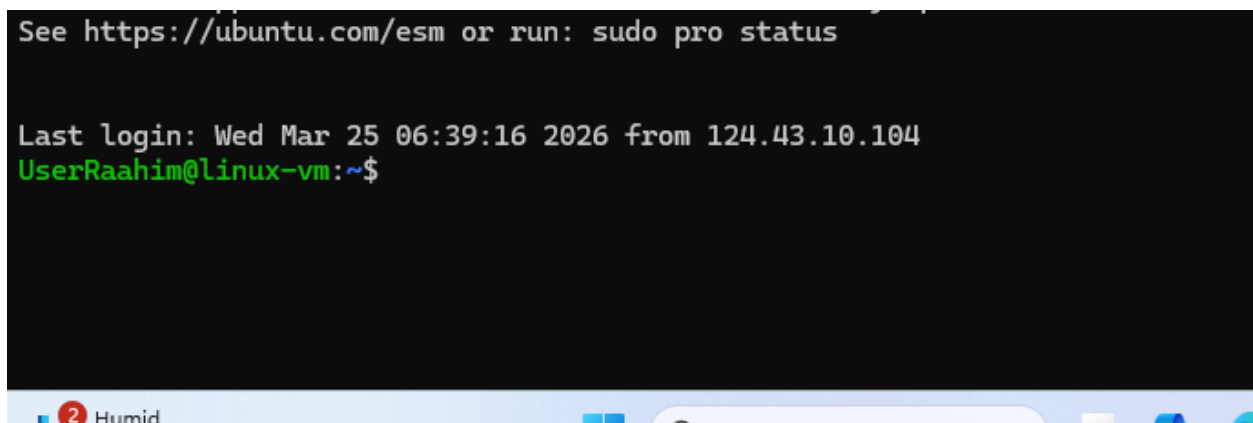
- Deployed PAM VM using a community image
- Automatically created:
 - Virtual Network (**vnet-uaenorth**)
 - Subnet and NSG

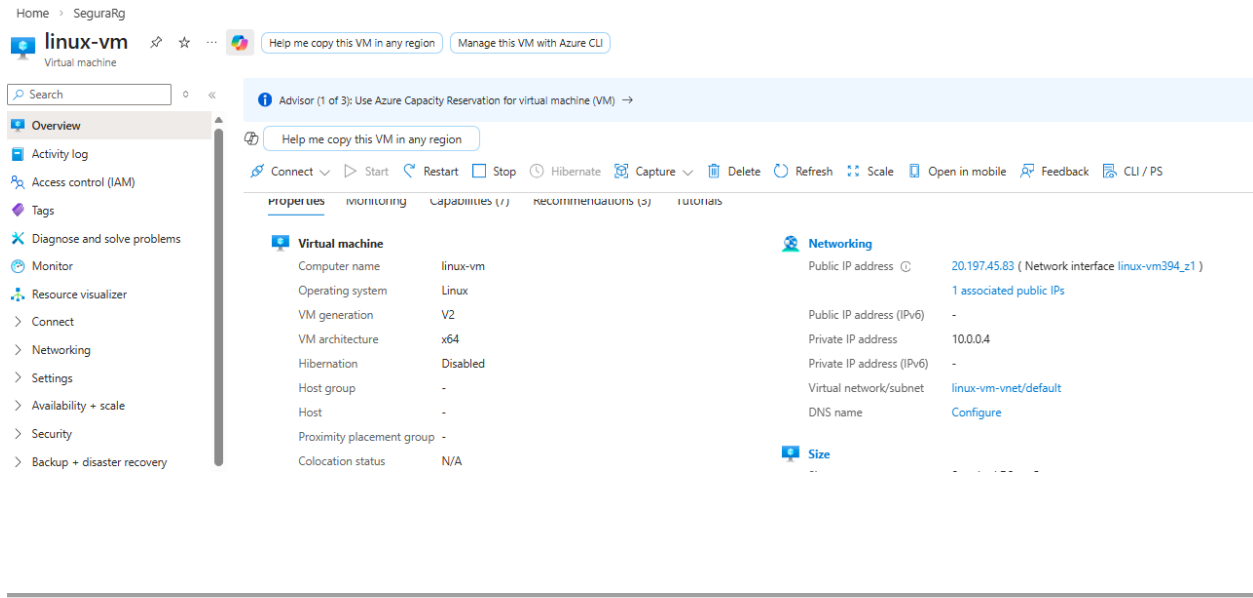


Step 2: Deploy Target VMs

◆ Linux VM

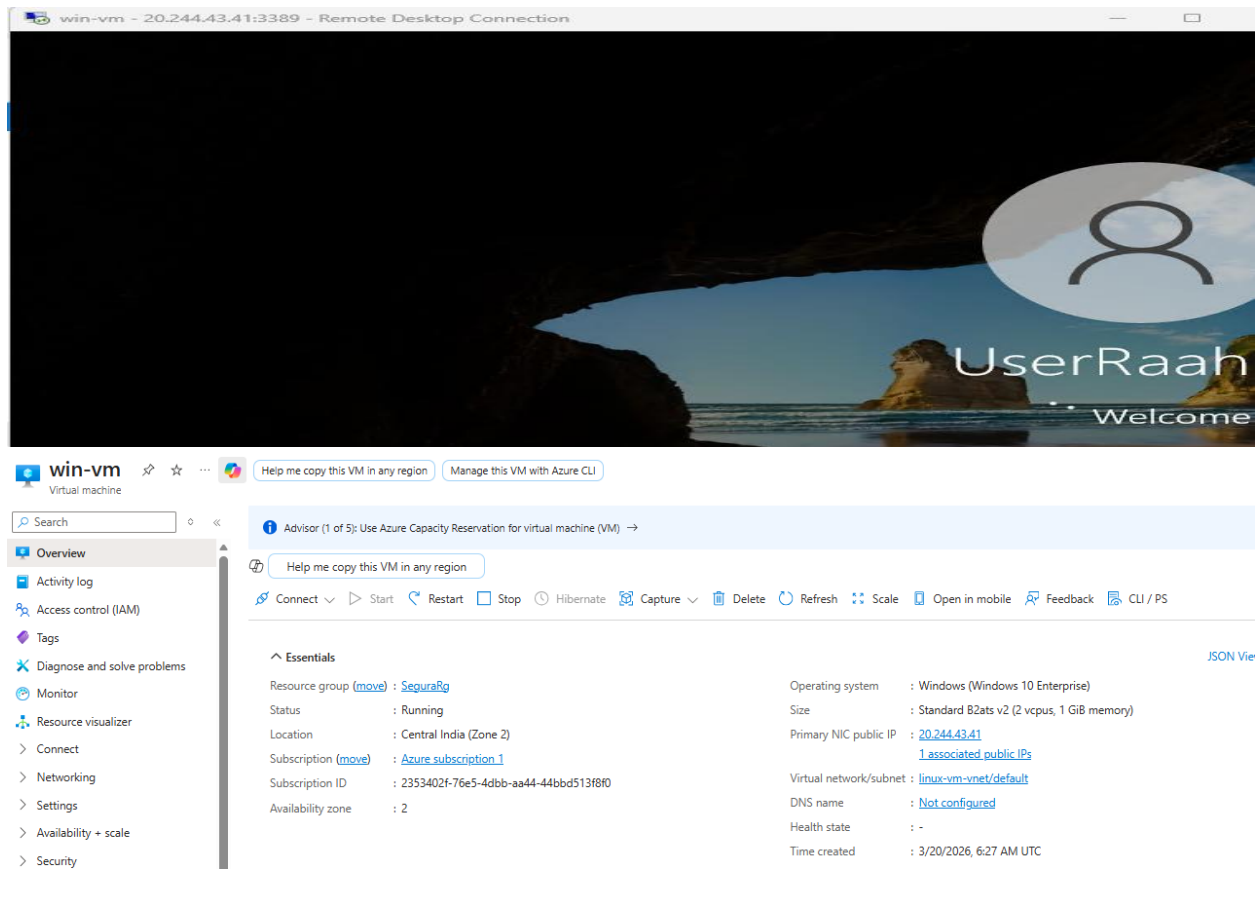
- Created in Central India
- Configured:
 - VNet: **linux-vm-vnet**
 - Address space: 10.0.0.0/16
 - Enabled SSH (Port 22)





◆ Windows VM

- Created in same VNet (**linux-vm-vnet**)
- Configured:
 - Enabled RDP (Port 3389)
 - Assigned local administrator credentials



⚠️ Challenge Faced

- Unable to deploy low-cost VMs in UAE North due to:
 - Region restrictions
 - Image limitations
 - Subscription quota limits

✅ Solution

- Deployed target systems in **Central India**
- Connected both VNets using **VNet Peering**

Step 3: Configure VNet Peering

Setting	Value
Peering Type	Bidirectional
Source VNet	vnet-uaenorth
Remote VNet	linux-vm-vnet
Allow VNet Access	 Enabled
Allow Forwarded Traffic	 Enabled
Gateway	 Disabled

Home

vnet-uaenorth

Virtual network

Search

0

«

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Resource visualizer

Settings

Monitoring

Automation

Help

→ Move

🗑️ Delete

🔄 Refresh

🗨️ Give feedback

Subscription [\(move\)](#)

: [Azure subscription 1](#)

Subscription ID

: 2353402f-76e5-4d8b-aa44-44bbd513f8f0

DNS servers

: [Azure provided DNS service](#)

BGP community string

: [Configure](#)

Virtual network ID

: 15a007df-f33f-46f7-9de1-6a1509955ffe

Tags [\(edit\)](#)

: [Add tags](#)

Topology

Properties

Capabilities (5)

Recommendations

Tutorials

🛡️ DDoS protection

Configure additional protection from distributed denial of service attacks.

● Not configured

🔥 Azure Firewall

Protect your network with a stateful L3-L7 firewall.

● Not configured

🌐 Peerings

Seamlessly connect two or more virtual networks.

🟢 1 peering

🛡️ Microsoft Defender for Cloud

Strengthen the security posture of your environment.

🔗 Private endpoints

Privately access Azure services

vnet-uaenorth | Peerings

Virtual network

Search

0

«

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Resource visualizer

Settings

Address space

Connected devices

+ Add

🔄 Refresh

📄 Export to CSV

🗑️ Delete

🔄 Sync

Virtual network peering enables you to seamlessly connect two or more virtual networks in Azure. The virtual networks appear as one for connectivity purposes. [Learn more](#)

Filter by name...

Showing all 1 items

☐	Name	Peering sync status	Peering state	Remote VNet	Virtual network	Cross-tenant
☐	india-to-us	🟢 Fully Synchronized	🟢 Connected	linux-vm...	Disabled	No

✓ Result

- Peering Status: **Connected**
- Sync Status: **Fully Synchronized**

🔑 Step 4: Configure Network Security Groups (NSG)

◆ Linux VM NSG

- Allow SSH:
 - Source: 172.16.0.0/16
 - Port: 22
- Allow ICMP (for testing)

Home > linux-vm

linux-vm | Network settings ☆ ... [How can I make this VM secure?](#) [What are the requirements for attaching and detaching network interfaces?](#) [List all my network interfaces for this VM](#)

Search

Overview
Activity log
Access control (IAM)
Tags
Diagnose and solve problems
Monitor
Resource visualizer
Connect
Networking
Network settings
Load balancing
Application security groups
Network manager

Search rules

Source == all Destination == all Protocol == all Action == all Port == all

Priority ↑	Name	Port	Protocol	Source	Destination	Action
Inbound port rules (5)						
100	allow_from_	Any	Any	172.16.0.0/16	Any	Allow
300	SSH	22	TCP	Any	Any	Allow
65000	AllowVnetInBound	Any	Any	VirtualNetwork	VirtualNetwork	Allow
65001	AllowAzureLoadBalancerInBound	Any	Any	AzureLoadBalancer	Any	Allow
65500	DenyAllInBound	Any	Any	Any	Any	Deny
Outbound port rules (4)						
100	allow_traffic_172	Any	Any	Any	172.16.0.0/16	Allow
65000	AllowVnetOutBound	Any	Any	VirtualNetwork	VirtualNetwork	Allow
65001	AllowInternetOutBound	Any	Any	Any	Internet	Allow

◆ Windows VM NSG

- Allow RDP:
 - Source: 172.16.0.0/16
 - Port: 3389

Home > win-vm

win-vm | Network settings ☆ ... [List all my network interfaces for this VM](#) [What are the requirements for attaching and detaching network interfaces?](#) [How can I make this VM secure?](#)

Virtual machine

Search

Overview
Activity log
Access control (IAM)
Tags
Diagnose and solve problems
Monitor
Resource visualizer
Connect
Networking

Search rules Source == all Destination == all Protocol == all Action == all Port == all

Priority ↑	Name	Port	Protocol	Source	Destination	Action
Inbound port rules (5)						
100	allow_from	Any	Any	172.16.0.0/16	Any	Allow
300	RDP	3389	TCP	Any	Any	Allow
65000	AllowVnetInBound	Any	Any	VirtualNetwork	VirtualNetwork	Allow
65001	AllowAzureLoadBalancerInBound	Any	Any	AzureLoadBalancer	Any	Allow
65500	DenyAllInBound	Any	Any	Any	Any	Deny
Outbound port rules (4)						

◆ PAM VM NSG

- Allowed ICMP from Linux subnet (10.0.0.0/16)
- Observed restricted inbound behavior

Home > MYPAM

MYPAM | Network settings ☆ ... [What are the requirements for attaching and detaching network interfaces?](#) [How can I make this VM secure?](#) [List all my network interfaces for this VM](#)

Virtual machine

Search

Overview
Activity log
Access control (IAM)
Tags
Diagnose and solve problems
Monitor
Resource visualizer
Connect
Networking
Network settings
Load balancing
Application security groups
Network manager
Settings

Search rules

Inbound port rules (8)						
100	allow_from_10.0.0	Any	Any	10.0.0.0/16	Any	Allow
290	ssh2	59022	TCP	Any	Any	Allow
300	SSH	22	TCP	Any	Any	Allow
320	HTTP	80	TCP	Any	Any	Allow
340	HTTPS	443	TCP	Any	Any	Allow
65000	AllowVnetInBound	Any	Any	VirtualNetwork	VirtualNetwork	Allow
65001	AllowAzureLoadBalancerInBound	Any	Any	AzureLoadBalancer	Any	Allow
65500	DenyAllInBound	Any	Any	Any	Any	Deny
Outbound port rules (4)						
100	allow_traffic_10.0	Any	Any	Any	10.0.0.0/16	Allow
65000	AllowVnetOutBound	Any	Any	VirtualNetwork	VirtualNetwork	Allow
65001	AllowInternetOutBound	Any	Any	Any	Internet	Allow
65500	DenyAllOutBound	Any	Any	Any	Any	Deny

Virtual appliance itself block all inbound traffic unless the specific port

🧠 Key Observation

- PAM Appliance:
 -  Allows outbound traffic
 -  Restricts inbound ICMP
- Target VMs:

- Require explicit NSG + firewall configuration
-

Step 5: Firewall Configuration

Linux:

`sudo ufw status`

`sudo ufw disable` # for testing

Windows:

- Enabled RDP in System Settings
 - Allowed RDP via Windows Defender Firewall
-

Step 6: Connectivity Testing

Test 1: PAM → Linux

`ping 10.0.0.x`

✓ Success

```
mt4adm@segura:~$  
mt4adm@segura:~$ ping 10.0.0.4  
PING 10.0.0.4 (10.0.0.4) 56(84) bytes of data.  
64 bytes from 10.0.0.4: icmp_seq=1 ttl=64 time=29.3 ms  
64 bytes from 10.0.0.4: icmp_seq=2 ttl=64 time=28.8 ms  
^[[064 bytes from 10.0.0.4: icmp_seq=3 ttl=64 time=28.8 ms
```

Test 2: Linux → PAM

`ping 172.16.0.x`

 Blocked (expected)

```
Last login: Wed Mar 25 19:41:42 2026 from 172.16.0.4
UserRaahim@linux-vm:~$ ping 172.16.0.4
PING 172.16.0.4 (172.16.0.4) 56(84) bytes of data.
```

✓ Test 3: SSH

ssh [user@10.0.0.x](#)

```
mt4adm@segura:~$ telnet 10.0.0.4 22
Trying 10.0.0.4...
Connected to 10.0.0.4.
Escape character is '^]'.
SSH-2.0-OpenSSH_9.6p1 Ubuntu-3ubuntu13.15
^[[0
```

✓ Success

✓ Successful

✓ Test 4: RDP

- Connected to Windows VM using RDP

✓ Successful

```
mt4adm@segura:~$ telnet 10.0.0.5 3389
Trying 10.0.0.5...
Connected to 10.0.0.5.
Escape character is '^]'.
^[[0
```

•  Management

❖ Step 7: PAM Integration – on bored you devises

- Added Linux & Windows devices into senhasegura
- Configured:
 - Linux → SSH (Port 22)
 - Windows → RDP (Port 3389)

Devices ⓘ
PAM Core - Devices - All devices Add

ID Name Management Status Enabled More

Filter Clear

ID	Name	Management	Type	Vendor	Product	Site	Connectivity	Domains	Tags	Status	Actions
1	linux-vm	10.0.0.4	Server	Ubuntu	Ubuntu Server 17.10	Default	SSH [22]: Connectable		lab, linux, testdevice	Enabled	
2	win-vm	20.244.43.41	Workstation	Microsoft	Windows 10	Default	RDP [3389]: Connectable		windows	Enabled	

Total: 2 records

30 records << < 1 of 1 > >>

• Map credentials

PAM Core Search (CTRL + G) segura Home + Alert ? Chat

Access credentials ⓘ
PAM Core - Credentials - All credentials Add

ID Device Username Status Enabled More

Filter Clear

ID	Type	Username	Additional	Domain	Tags	Session status	Actions
linux-vm (10.0.0.4): Server Ubuntu Server 17.10 Default 1							
11	Local User	UserRaahim				Available	
win-vm (20.244.43.41): Workstation Windows 10 Default 2							
10	Local administrator	UserRaahim				Available	

Total: 2 records

30 records << < 1 of 1 > >>

⚠ Critical Issue Faced (Advanced Troubleshooting)

- PAM sessions failed despite:
 - Successful authentication
 - Working SSH/RDP manually

Root Cause:

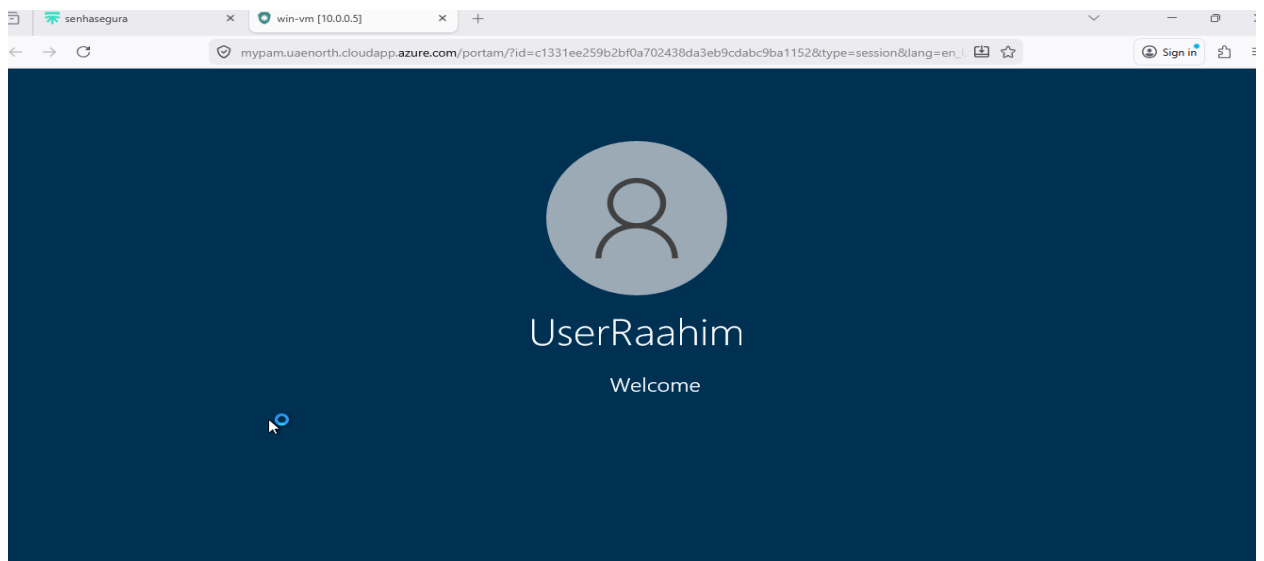
✗ Terminal Proxy container failed due to port 22 conflict : The PAM system successfully authenticated the user and initiated a session, but failed to establish the backend connection to the target machine via Terminal Proxy / RDP Proxy, resulting in session failure.

- Host SSH service already using port 22
- Proxy container unable to bind → crashed

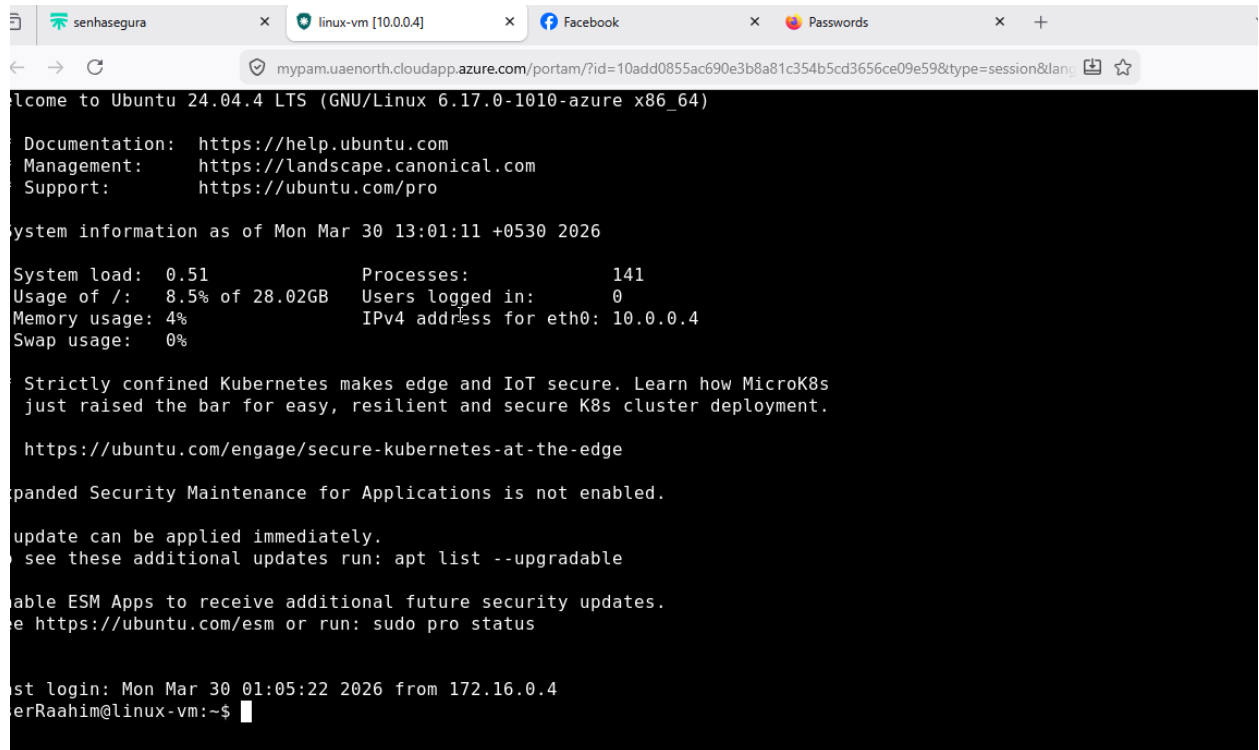
Fix Implemented

- Modified proxy configuration:
 - Changed port mapping → **2222:22**
- Restarted proxy services
- SSH sessions via PAM worked successfully

windows



Linux VM



The screenshot shows a terminal window titled 'linux-vm [10.0.0.4]' within a browser. The terminal output displays the Ubuntu 24.04.4 LTS boot sequence, including system load, memory usage, and network configuration. The user 'erRaahim' is logged in from IP 172.16.0.4.

```
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-1010-azure x86_64)

Documentation:  https://help.ubuntu.com
Management:    https://landscape.canonical.com
Support:        https://ubuntu.com/pro

System information as of Mon Mar 30 13:01:11 +0530 2026

System load:  0.51               Processes:            141
Usage of /:    8.5% of 28.02GB   Users logged in:     0
Memory usage:  4%               IPv4 address for eth0: 10.0.0.4
Swap usage:    0%

Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
just raised the bar for easy, resilient and secure K8s cluster deployment.

https://ubuntu.com/engage/secure-kubernetes-at-the-edge

Expanded Security Maintenance for Applications is not enabled.

No new update can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Mon Mar 30 01:05:22 2026 from 172.16.0.4
erRaahim@linux-vm:~$
```

💡 Key Learnings

- VNet Peering enables secure cross-region communication
- ICMP does not use ports
- NSG rules must be configured on both ends
- PAM introduces a **proxy layer (Terminal/RDP Proxy)**
- Port conflicts can break session handling
- Troubleshooting requires:
 - Logs
 - Docker containers
 - System services

Cost Optimization Strategy

- Used low-cost VM sizes (B-series)
- Avoided VPN Gateway
- Used VNet Peering (cost-efficient)
- Deallocated VMs when idle

Conclusion

This lab successfully demonstrated:

- ✓ Cross-region cloud networking
- ✓ Secure privileged access using PAM
- ✓ Integration of Linux (SSH) and Windows (RDP) systems
- ✓ Real-world troubleshooting of proxy and port conflicts
- ✓ Cost-efficient cloud architecture design